Handling Data 1 — Types of data · Presenting data · Averages

Calculator allowed. Squared paper and a protractor for the drawing questions. Marks in brackets. ★ = harder question.

1 Say whether each type of data is discrete, continuous or categorical: **a** the number of goals scored in a match, **b** the height of a plant, **c** a person's favourite colour, **d** shoe size. [2]

> **a** Discrete (counted). **b** Continuous (measured — any value). **c** Categorical (a label, not a number). **d** Discrete (fixed sizes).

2 A pie chart shows how the 48 students in a year group travel to school. The sector angles are: bus $120°$, walk $135°$, car $60°$, bicycle $45°$. [3]

a How many students come by bus?

b Next year 9 students cycle, out of the same 48. What should the bicycle sector's angle be?

> **a** $\dfrac{120}{360} \times 48 = 16$ students.
>
> **b** $\dfrac{9}{48} \times 360 = 67.5°$.

3 All 180 students in Year 11 were asked whether they walk or cycle to school. 125 said they walk, 50 said they cycle, and 15 do neither. How many both walk and cycle? [2]

> Walk or cycle (or both) $= 180 - 15 = 165$. Walk + cycle $= 175$ counts the "both" students twice: both $= 175 - 165 = 10$.

4 The table shows the medals won by a team in two years. [5]

	Gold	Silver	Bronze
Year 1	12	18	15
Year 2	26	17	20

a Draw a comparative (dual) bar chart to show this data.

b Comment on the difference between the two years.

> **a** Three groups (Gold, Silver, Bronze), each with two bars side by side (Year 1, Year 2), a key, frequency axis from 0 to 30 in 5s.
>
> **b** Year 2 was better overall (63 medals vs 45). The big change is gold: more than doubled ($12 \to 26$); silver stayed about the same, bronze rose a little.

5 A café serves a set menu: soup or salad to start, then fish or vegetable curry. One lunchtime there were 60 customers. Copy and complete the two-way table. [3]

	Fish	Vegetable	Total
Soup	18		27
Salad		8	
Total			60

> Soup–vegetable $= 27 - 18 = 9$. Salad total $= 60 - 27 = 33$; salad–fish $= 33 - 8 = 25$. Fish total $= 18 + 25 = 43$; vegetable total $= 9 + 8 = 17$ (check: $43 + 17 = 60$ ✓).

6 A drinks company's bar chart claims cola sells "more than twice as much" as other drinks. The vertical axis starts at 300 (not 0), and the cola bar is drawn wider than the others. Give two reasons why the chart is misleading. [2]

> Starting the axis at 300 exaggerates the differences — a bar of 480 looks several times taller than one of 380 although it is only 26% more. The wider bar makes cola look bigger still; bar *height* should be the only thing that varies. (Also acceptable: no units, no scale on the axis.)

7 Find the mean, mode and median of $3,\ 8,\ 5,\ 12,\ 8,\ 0,\ 6,\ 9,\ 3$. [3]

> Mean $= \dfrac{54}{9} = 6$. In order: $0, 3, 3, 5, 6, 8, 8, 9, 12$ → median (5th of 9) $= 6$. Mode: 3 and 8 both appear twice — two modes.

8 To get a distinction, Eleni's mean mark over five tests must be at least 85. Her first four marks are 86, 94, 75 and 88. What is the lowest mark she can get in the fifth test? [3]

> Total needed $= 85 \times 5 = 425$. So far $86 + 94 + 75 + 88 = 343$. Fifth test $\geq 425 - 343 = 82$.

9 In a town, the mean number of parcels delivered per day was 2 in March and 1.6 in April. What was the mean number per day over the two months together? Give your answer to 3 s.f. [4]

> You cannot average the two means (the months have different lengths). March: $2 \times 31 = 62$ parcels; April: $1.6 \times 30 = 48$. Total 110 over 61 days: $\dfrac{110}{61} = 1.803\ldots \approx 1.80$ per day.

Total: 27 marks

Check: pie chart angle = fraction × 360° · a two-way table's rows and columns must each add to their totals · mean needs the total and the count — never average averages.